

Cambridge (CIE) IGCSE Chemistry
Paper 2: Multiple Choice (Set B)
Extended

Saturday 26 April 2025

Morning (Time: 0 hours 45 minutes)

Total marks

/ 40

Instructions

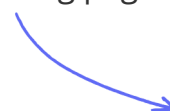
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.

Materials

- A copy of [the Periodic Table of elements](#) is required for this paper.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



1 The following four statements describe the arrangement of particles in a substance.

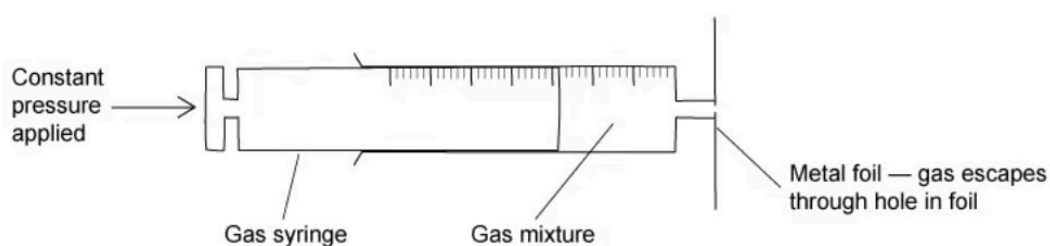
1	Particles are moving in all directions
2	Particles are arranged in neat rows
3	Particles can move past each other
4	Particles vibrate about their positions

Which of the four statements describe the arrangement of particles in a solid?

- A. 1 and 2
- B. 2 and 4
- C. 1 and 3
- D. 2 only

(1 mark)

2 The rate of diffusion of two gases, methane, CH_4 , and propene, C_3H_6 , is measured using the apparatus shown.



Which gas diffuses at a faster rate and why?

	Fastest gas	Reason
A	Propene	Propene molecules are heavier and so move faster.
B	Methane	Methane is a lighter molecule.
C	Propene	Propene is unsaturated so is more reactive.
D	Methane	Methane molecules are smaller so fit through the hole more easily.

(1 mark)

3 Calcium reacts with fluorine to form calcium fluoride

Which statement describes what happens to calcium atoms in this reaction?

- A.** Calcium ions lose electrons to form positive charges. .
- B.** Calcium atoms gain 3 electrons to form positive ions.
- C.** Calcium ions gain 2 electrons to form positive ions.
- D.** Calcium atoms lose 2 electrons to form positive ions.

(1 mark)

4 Which of the following statements about graphite and diamond are incorrect?

1	They are allotropes of carbon.
2	They both conduct electricity.
3	They form different numbers of bonds within their structures.
4	They have different uses.

- A.** 2 and 3
- B.** 2 only
- C.** 1, 3 and 4
- D.** 2 and 4

(1 mark)

5 The following statements describe the structure and properties of silicon(IV) oxide.

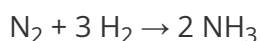
1. Giant covalent structure held together by many strong covalent bonds.
2. It is a very hard substance.

Which of the following is correct for statements 1 and 2?

- A.** Statement 2 is correct but statement 1 is incorrect.
- B.** Both statements are incorrect.
- C.** Both are correct and statement 1 explains statement 2.
- D.** Both are correct and statement 2 explains statement 1.

(1 mark)

6 Ammonia is manufactured by the reaction of hydrogen and nitrogen in the Haber process, as shown in the equation below:



16.8 tonnes of ammonia can be produced from 14 tonnes of nitrogen.

How much nitrogen is needed to produce 51 tonnes of ammonia?

- A.** 42.5
- B.** 34
- C.** 17
- D.** 24.5

(1 mark)

7 The structure of four particles is described in the table.

Particle	Number of protons	Number of neutrons	Number of electrons
Li	3	X	3
Li ⁺	3	4	2
F	Y	10	9
F ⁻	9	10	Z

What are the correct values for X, Y and Z?

	X	Y	Z
A	4	7	9
B	3	9	9
C	4	9	10
D	3	7	10

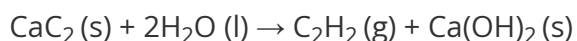
(1 mark)

8 What is the concentration in mol dm^{-3} of a solution of sodium hydroxide that contains 16 g of NaOH in 200 cm^3 of distilled water?

- A. 2 mol dm^{-3}
- B. 2.5 mol cm^{-3}
- C. 0.2 mol dm^{-3}
- D. 1 mol cm^{-3}

(1 mark)

9 Calcium carbide and water react to produce ethyne and calcium hydroxide.



What is the volume of C_2H_2 produced, at standard temperature and pressure, when 45 g of water reacts completely with calcium carbide?

- A. 72 dm^3
- B. 30 dm^3
- C. 24 dm^3
- D. 6 dm^3

(1 mark)

10 Dilute sulfuric acid is electrolysed using inert electrodes.

What are the products of this electrolysis?

	cathode	anode
A	oxygen	sulfur
B	hydrogen	sulfur dioxide
C	oxygen	hydrogen
D	hydrogen	oxygen

(1 mark)

11 Which metal could **not** be used for electroplating by using an aqueous solution?

- A.** Chromium
- B.** Copper
- C.** Silver
- D.** Sodium

(1 mark)

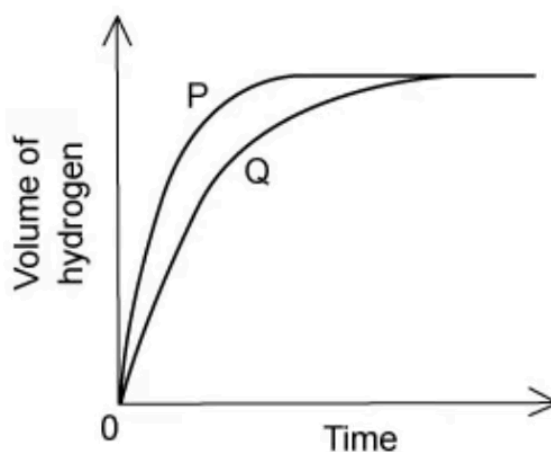
12 In all chemical reactions, bonds in the reactant species are broken and new bonds are formed in the products. Which statement is correct?

- A.** Bond breaking requires energy and is thus an exothermic process.
- B.** Bond breaking requires energy and is thus an endothermic process.
- C.** Bond breaking releases energy and is thus an exothermic process.
- D.** Bond breaking releases energy and is thus an endothermic process.

(1 mark)

13 A student was investigating the rate of reaction between iron and a solution of sulfuric acid. Two experiments were performed, P and Q, in which the volume of hydrogen was

recorded at regular intervals.



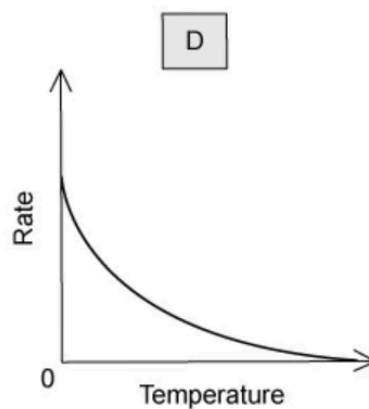
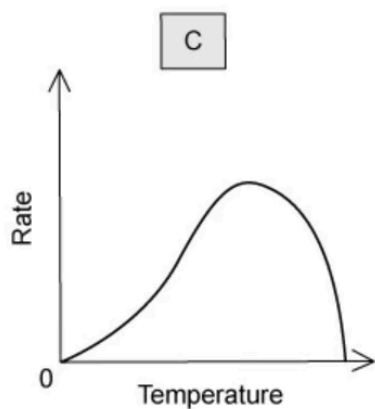
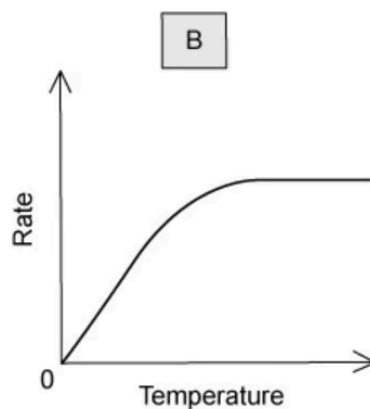
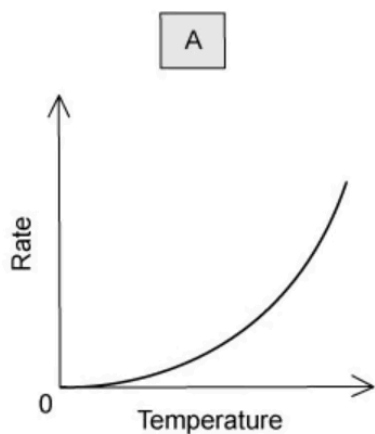
Which change could explain the difference in results between P and Q?

- A.** The student added a catalyst in P.
- B.** Larger pieces of iron were used in P.
- C.** Less concentrated acid was used in P.
- D.** Larger quantities of both reactants are used P.

(1 mark)

14 Which graph correctly shows the effect of increasing the temperature on the rate of

reaction between magnesium and sulfuric acid in which neither reactant runs out?

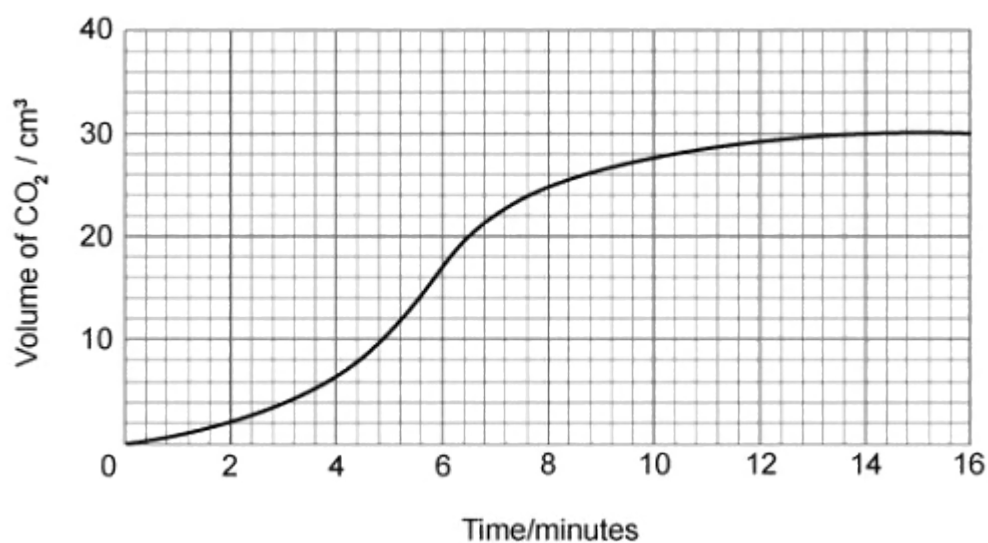


- A.** Option A
- B.** Option B
- C.** Option C
- D.** Option D

(1 mark)

- 15** A student reacts a metal carbonate with nitric acid and measures the volume of carbon dioxide gas released at regular intervals. The results of the experiment are shown in the

graph.

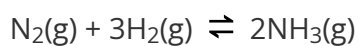


Between which two measurements was the reaction the fastest?

- A.** 0 and 2 minutes
- B.** 2 and 4 minutes
- C.** 4 and 6 minutes
- D.** 10 and 12 minutes

(1 mark)

16 The reaction for the production of ammonia is shown below.



What is the effect of increasing pressure on the rate of production of ammonia and the percentage yield of ammonia?

	rate of production	percentage yield
A	increases	decreases
B	increases	increases
C	decreases	increases
D	decreases	decreases

(1 mark)

17 Three reactions are shown below.

- $\text{NO} + \text{O}_3 \rightarrow \text{NO}_2 + \text{O}_2$
- $2\text{C} + \text{O}_2 \rightarrow 2\text{CO}$
- $2\text{MgO} + \text{O}_2 \rightarrow 2\text{MgO}_2$

Which row correctly describes the process each reactant molecule undergoes?

	NO	O ₃	MgO
A	oxidised	reduced	oxidised
B	oxidised	oxidised	reduced
C	reduced	oxidised	oxidised
D	reduced	reduced	oxidised

(1 mark)

- 18 Most insect bites cause a stinging sensation because of the injection of a weak acid into the surface of the skin. Which substance could be used to neutralise the acid from an insect bite?

	substance	pH
A	battery acid	1
B	toothpaste	8
C	grill cleaner	13
D	lime juice	3

(1 mark)

- 19 A student is provided with an unmarked beaker containing a colourless solution and asked to identify it. The student decides to perform a series of reactions with the solution and record the observations.

Which of these observations is not indicative of an acid?

- A. The solution causes blue litmus to turn red.
- B. On heating and the addition of copper oxide, the solution turns blue.
- C. On the addition of ammonium carbonate, a colourless, pungent gas is released.
- D. On the addition of a piece of magnesium, effervescence was produced and the magnesium piece 'disappeared' after a few minutes.

(1 mark)

- 20 Bromine and rubidium react together to form the compound rubidium bromide. Which row correctly describes the formula and colour of rubidium bromide?

	formula	colour
A	RbBr	dark grey
B	RbBr ₂	dark grey
C	RbBr	white
D	RbBr ₂	white

(1 mark)

- 21** Astatine is a radioactive Group VII element which is extremely unstable, making it difficult to study. The physical and chemical properties of astatine can only be estimated.

Which row most likely describes the properties of astatine?

	State	Colour	Reaction with KI (aq)
A	liquid	yellow	brown colour
B	solid	green	no reaction
C	gas	dark brown	brown colour
D	solid	black	no reaction

(1 mark)

- 22** A solution of a lithium halide and a halogen are mixed together. A reaction occurs and the mixture darkens in colour.

What is present in the reaction mixture?

- A.** Bromine and lithium chloride
- B.** Bromine and lithium fluoride
- C.** Chlorine and lithium iodide
- D.** Chlorine and lithium fluoride

(1 mark)

- 23** Astatine is found below iodine in Group 7 of the periodic table.

What is its formula and state at room temperature?

	Formula	State at room temperature
A	At	gas
B	At	solid
C	At ₂	liquid
D	At ₂	solid

(1 mark)

- 24** Brass is an alloy of two metals.

Which row describes the correct use of the two metals that brass is made from?

	metal 1	metal 2
A	galvanising steel	making aircraft
B	electrical wiring	galvanising steel
C	making aircraft	making cutlery
D	making cooking pans	electrical wiring

A. Option A

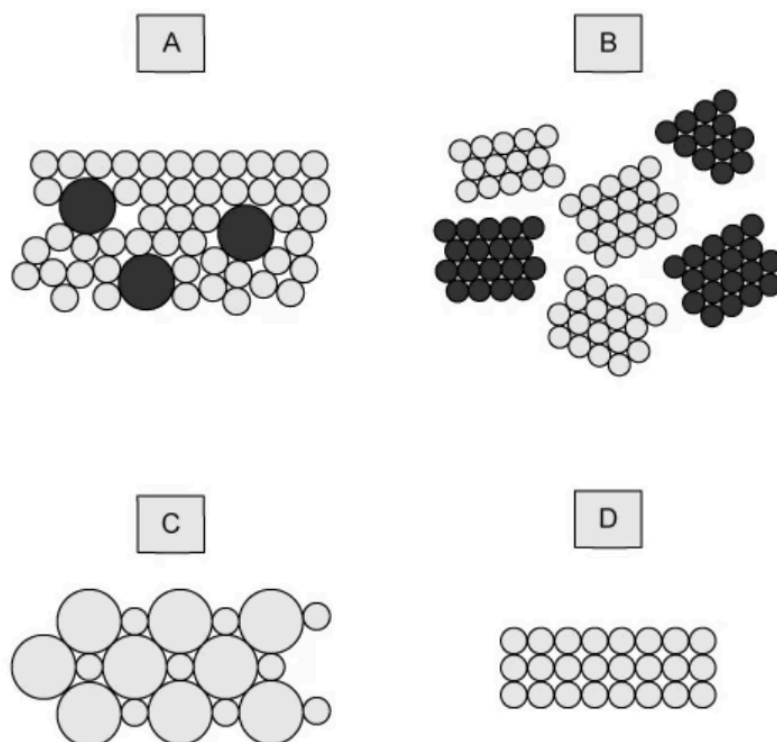
B. Option B

C. Option C

D. Option D

(1 mark)

25 Which diagram shows an alloy?



(1 mark)

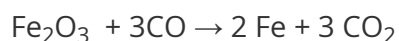
26 Small blocks of zinc are attached to ship hulls which are commonly made from steel, an alloy of iron. This is done to prevent the hull from rusting.

Which statement explains this process?

- A.** Zinc is higher up on the reactivity series than iron so has a higher tendency to become reduced.
- B.** Zinc is lower down on the reactivity series than iron so has a higher tendency to become reduced.
- C.** Zinc is higher up on the reactivity series than iron so has a higher tendency to become oxidised.
- D.** Zinc is lower down on the reactivity series than iron so has a higher tendency to become oxidised.

(1 mark)

- 27** In a blast furnace, iron (III) oxide is converted into iron and carbon monoxide is converted into carbon dioxide, as shown in the equation:

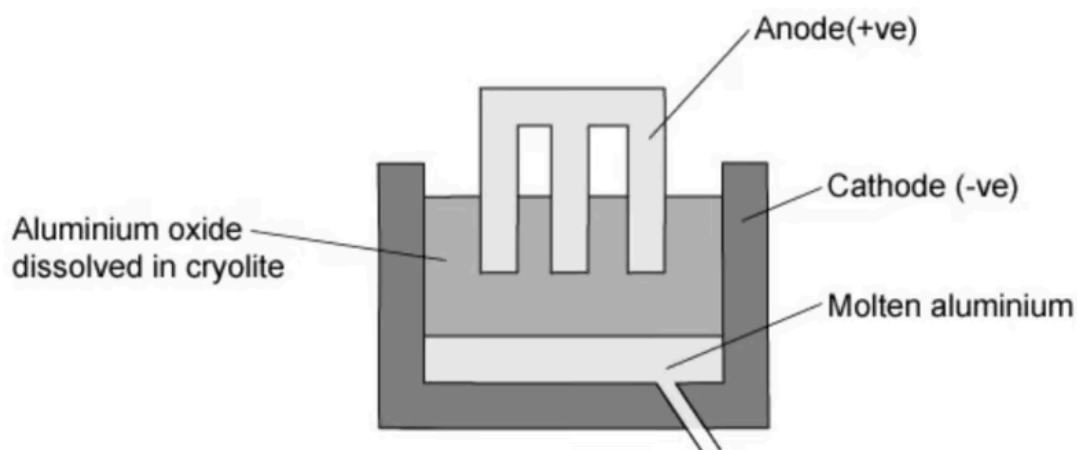


What happens to each of these reactants?

- A.** Iron (III) oxide is reduced and carbon monoxide is oxidised.
- B.** Iron (III) oxide is oxidised and carbon monoxide is reduced.
- C.** Both iron (III) oxide and carbon monoxide are oxidised.
- D.** Both iron (III) oxide and carbon monoxide are reduced.

(1 mark)

- 28** The diagram shows how aluminium is manufactured by electrolysis.



What are the anode and cathode made of?

	anode	cathode
A	aluminium	aluminium
B	aluminium	graphite
C	graphite	aluminium
D	graphite	graphite

A. Option A

B. Option B

C. Option C

D. Option D

(1 mark)

29 Reactions of three metals and their oxides are shown in the table.

metal	reacts with cold water	metal oxide reacts with carbon
P	no	no
Q	no	yes
R	yes	no

What is the order of reactivity of the metals?

	Least reactive	→	Most reactive
A	P	Q	R
B	Q	P	R
C	Q	R	P
D	R	P	Q

A. Option A

B. Option B

C. Option C

D. Option D

(1 mark)

30 Tin is less reactive than iron and the metal is extracted from its ore cassiterite, SnO_2 .

Which of the following statements are correct?

1. Tin does not conduct electricity.
2. Tin is hard and shiny
3. Tn can be extracted from cassiterite using carbon.

A. 1, 2 & 3

B. 1 & 2 only

C. 1 & 3 only

D. 2 & 3 only

(1 mark)

31 Which row correctly shows two ways of reducing the formation of acid rain?

A	Use a catalytic convertor	Reduce the amount of livestock farming
B	Plant more trees	Use fuels containing low levels of sulfur
C	Use fuels containing low levels of sulfur	Use a catalytic convertor
D	Reduce the amount of livestock farming	Plant more trees

(1 mark)

32 Flue gas desulfurisation uses calcium oxide to reduce the emissions of sulfur dioxide.

Which statements are correct?

1	Sulfur dioxide is a basic oxide
2	Sulfur dioxide is responsible for the production of acid rain
3	Calcium oxide reacts with acids
4	Sulfur dioxide emissions can also be reduced by using low sulfur fuels

A. 1, 2 and 3

B. 1 and 2 only

C. 1, 3 and 4

D. 2, 3 and 4

(1 mark)

33 Four chemical processes are described below. Which reactions contribute to global warming?

1	Reacting hydrochloric acid with magnesium carbonate.
2	The complete combustion of propane.
3	Heating calcium carbonate.
4	Reacting sodium hydroxide with magnesium carbonate.

A. 1, 2 & 3

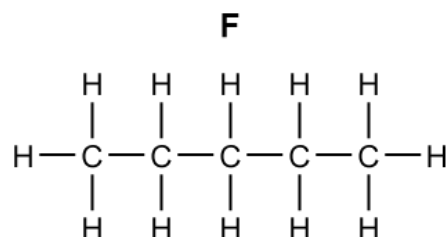
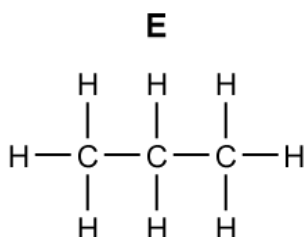
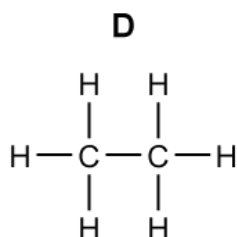
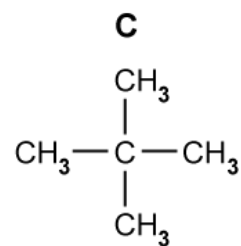
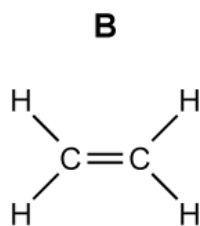
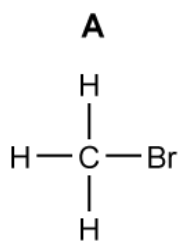
B. 1 & 4

C. 1, 3 & 4

D. All 4

(1 mark)

34 The structures of six organic compounds compound **A** are shown.



Which two compounds are isomers of one another?

A. B and D

B. A and C

C. C and F

D. C and E

(1 mark)

- 35** The table below shows 4 different sources of petroleum (crude oil) and their composition.

Fraction	Arabian Heavy /%	Arabian Light /%	Iranian Heavy /%	North Sea /%
gasoline	18	21	21	23
kerosene	11.5	13	13	15
diesel	18	20	20	24
fuel oil	52.5	46	46	38

Which composition of petroleum would be the best for the motor vehicle industry?

A. North Sea

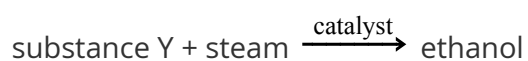
B. Arabian Heavy

C. Arabian Light

D. Iranian Heavy

(1 mark)

- 36** Ethanol can be produced from substance Y.



What is substance Y?

- A.** Ethane
- B.** Ethene
- C.** Carbon dioxide
- D.** Hydrogen

(1 mark)

37 The products of some reactions are shown below.

Which row correctly identifies the products of a reaction between a carboxylic acid and an alcohol?

- A.** Alkene and water
- B.** Alkane and water
- C.** Alcohol and water
- D.** Ester and water

(1 mark)

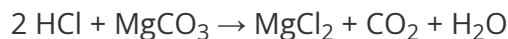
38 Which statement about polymers is correct?

- A.** Addition polymers are all biodegradable.
- B.** Condensation polymers can all be hydrolysed to give amino acids.
- C.** Condensation polymers only exist in nature.
- D.** Forming addition polymers produces only one product.

(1 mark)

- 39** A student wants to investigate the rate of reaction between marble chips and hydrochloric acid by measuring the volume of carbon dioxide produced with different concentrations of acid.

The equation for the reaction is as follows:



Which of the following lists of equipment can she use to measure the volume of the gas produced?

- A.** Gas syringe, cotton wool, bung, delivery tube
- B.** Gas syringe, conical flask, bung, delivery tube
- C.** Gas syringe, beaker, bung, balance
- D.** Balloon, conical flask, stopper, balance

(1 mark)

- 40** A student decided to investigate the rate of reaction between marble chips and dilute hydrochloric acid at 35 °C.

She had access to the following apparatus:

W a balance

X a thermometer

Y a measuring cylinder

Z a stopwatch

Which pieces of apparatus did she use?

- A.** **X**, **Y** and **Z** only
- B.** **W**, **Y** and **Z** only
- C.** **W**, **X** and **Z** only
- D.** **W**, **X**, **Y** and **Z**

(1 mark)